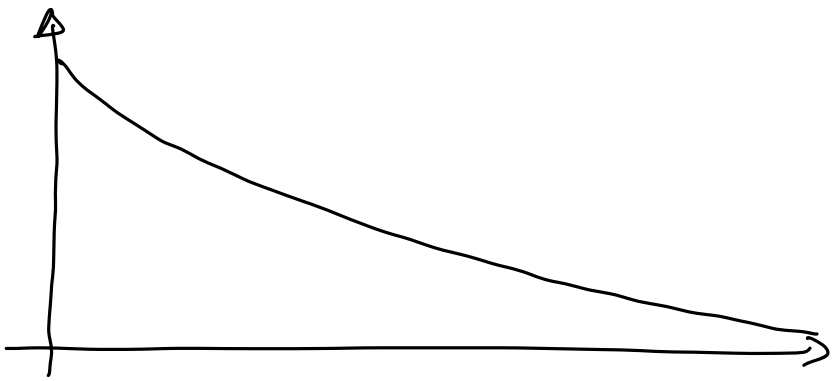
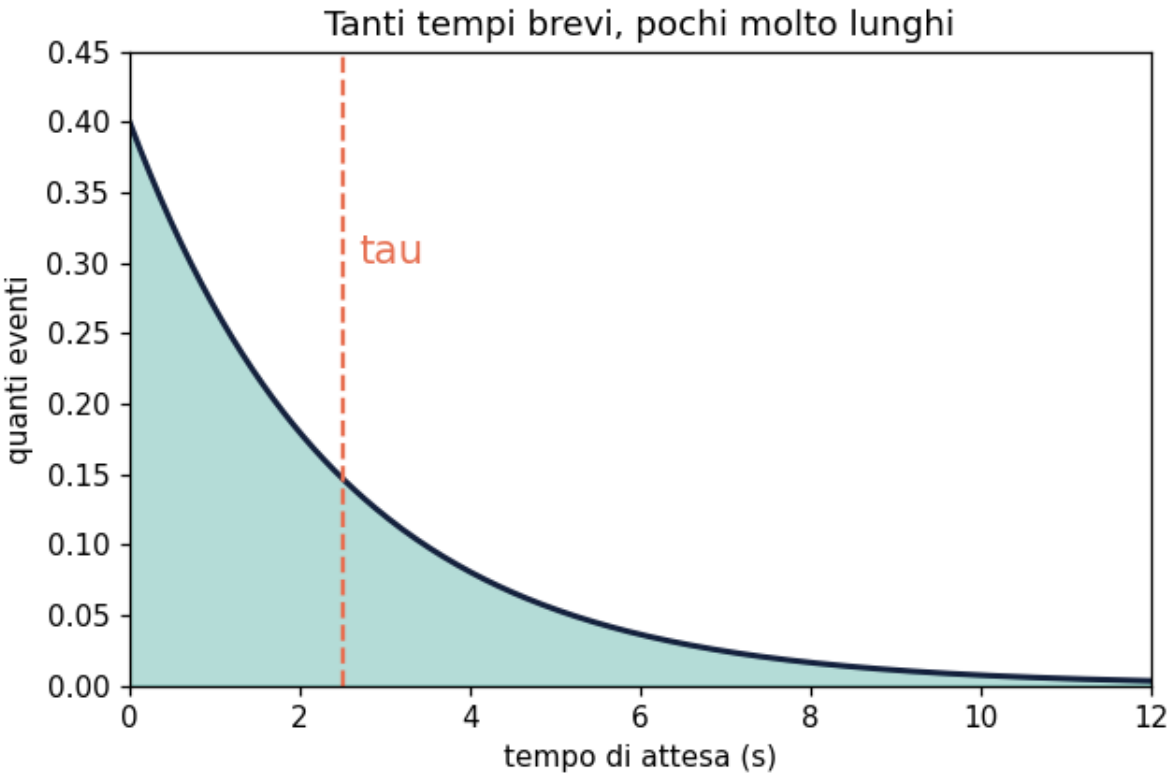
Distribuzione esponenziale

$$f(t) = \frac{e^{-t/\tau}}{\tau}$$



$$= \frac{e^{-\lambda t}}{\tau}$$

$$\lambda = \frac{1}{\tau}$$



esponenziale.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
#include <time.h>
#define N 100000
#define TAU 2.5      // tempo medio di attesa (s)

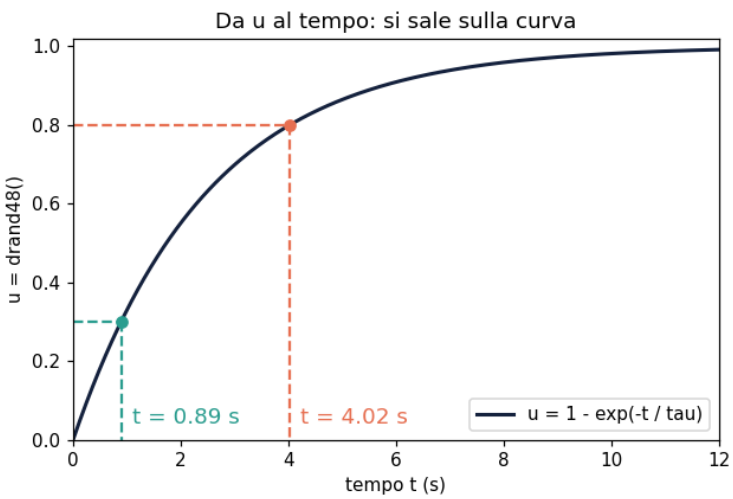
int main() {
    FILE *fp;
    double u, t, somma = 0.;
    int i, oltre = 0;
    srand48(time(NULL));
    fp = fopen("arrivi.txt", "w");
    if (fp == NULL) {
        printf("errore: impossibile aprire arrivi.txt\n");
        exit(-1);
    }
    for (i = 0; i < N; i++) {
        u = drand48();
        t = -TAU * log(1. - u);    // l'esponenziale
        fprintf(fp, "%.6f\n", t);
        somma += t;
        if (t > TAU) {
            oltre++;
        }
    }
    fclose(fp);
    printf("media dei tempi: %.4f\n", somma / N);
    printf("frazione oltre tau: %.4f\n", (double) oltre / N);
}
```

$u \in [0,1)$

$t = -\tau * \log(1-u)$

$u \in [0,1)$

$1-u \in (0,1]$



Terminale: esponenziale.c

```
$ gcc -Wall -Wextra -o app.exe esponenziale.c -lm
$ ./app.exe
media dei tempi: 2.4993
frazione oltre tau: 0.3683
$ wc -l arrivi.txt
  100000 arrivi.txt
```

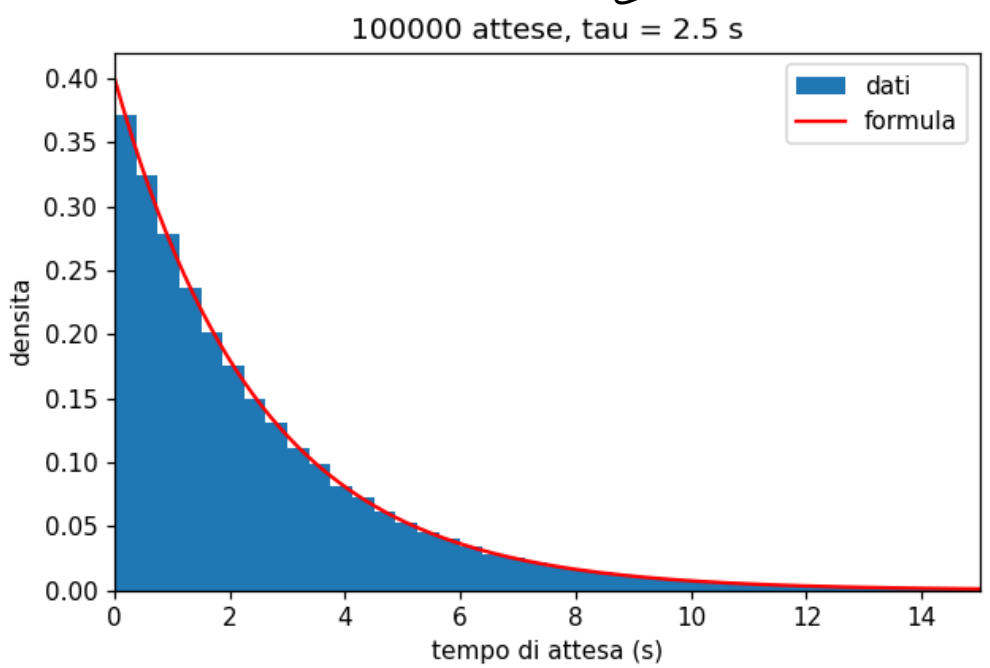
esponenziale.py

```python
import numpy as np
import matplotlib.pyplot as plt

TAU = 2.5
t = np.loadtxt('arrivi.txt')

plt.hist(t, bins=40, range=(0., 15.), density=True, label='dati')
tt = np.linspace(0., 15., 200)
f = np.exp(-tt / TAU) / TAU
plt.plot(tt, f, 'r-', label='formula')
plt.xlim(0., 15.)
plt.xlabel('tempo di attesa (s)')
plt.ylabel('densita')
plt.title('100000 attese, tau = 2.5 s')
plt.legend()
plt.savefig('esponenziale.png')
plt.show()
```

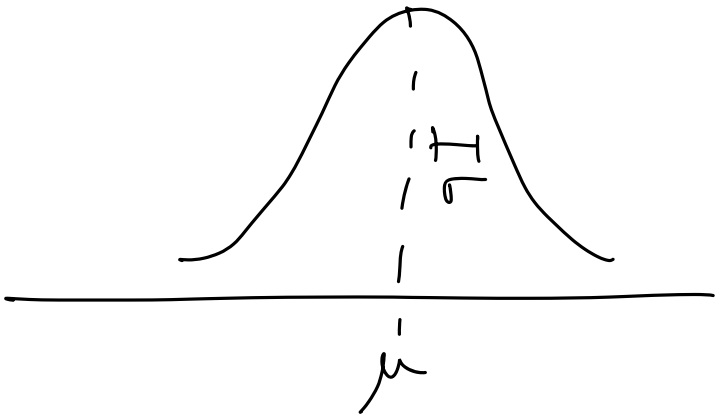
τ



intervallo range (0., 15.)

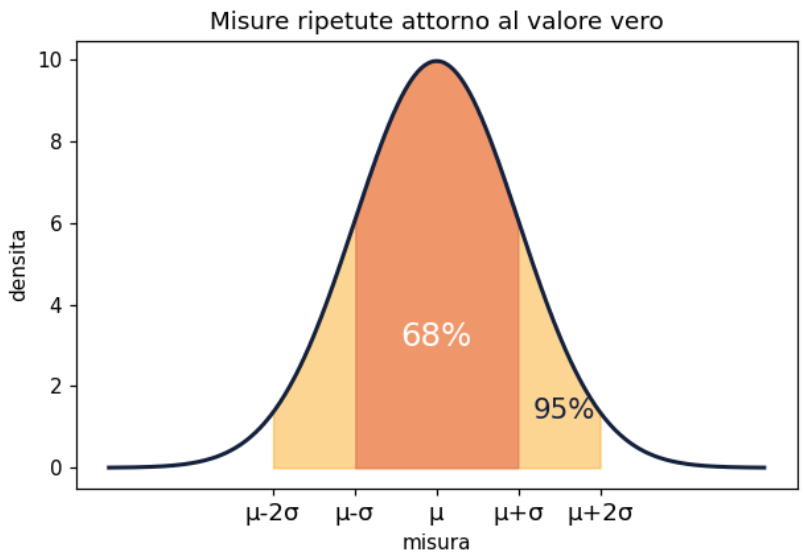
bins = 40 => larghezza intervallo $\frac{15}{40}$

Distribuzione Gaussiana



strumento funzionante senza problemi
genere misure gaussiane





$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

la ricetta (Box–Muller)

```
u1 = drand48();
u2 = drand48();
z = sqrt(-2. * log(1. - u1)) * cos(2. * M_PI * u2);
x = MU + SIGMA * z;
```

$u_1 \in [0,1)$
$u_2 \in [0,1)$
$1-u_1 \in (0,1]$

u1, u2 uniformi in [0,1) => 1 valore x gaussiano intorno a μ

gaussiana.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
#include <time.h>
#define N 100000
#define MU 25.00      // valore vero (mm)
#define SIGMA 0.04    // errore della misura (mm)

int main() {
    FILE *fp;
    double u1, u2, z, x;
    int i, entro1 = 0, entro2 = 0;
    srand48(time(NULL));
    fp = fopen("misure.txt", "w");
    if (fp == NULL) {
        printf("errore: impossibile aprire misure.txt\n");
        exit(-1);
    }
    for (i = 0; i < N; i++) {
        u1 = drand48();
        u2 = drand48();
        z = sqrt(-2. * log(1. - u1)) * cos(2. * M_PI * u2);
        x = MU + SIGMA * z;
        fprintf(fp, "%.6f\n", x);
        if (fabs(x - MU) < SIGMA) {
            entro1++;
        }
        if (fabs(x - MU) < 2. * SIGMA) {
            entro2++;
        }
    }
    fclose(fp);
    printf("entro 1 sigma: %.1f%%\n", 100. * entro1 / N);
    printf("entro 2 sigma: %.1f%%\n", 100. * entro2 / N);
}
```

Terminale: gaussiana.c

```
$ gcc -Wall -Wextra -o app.exe gaussiana.c -lm
$ ./app.exe
entro 1 sigma: 68.2%
entro 2 sigma: 95.4%
$ wc -l misure.txt
  100000 misure.txt
```

$ \sigma $

$ \mu $

gaussiana.py

```python
import numpy as np
import matplotlib.pyplot as plt

MU = 25.00
SIGMA = 0.04
x = np.loadtxt('misure.txt')

plt.hist(x, bins=40, density=True, label='dati')
xx = np.linspace(MU - 4. * SIGMA, MU + 4. * SIGMA, 200)
f = np.exp(-(xx - MU)**2 / (2. * SIGMA**2))
f = f / (SIGMA * np.sqrt(2. * np.pi))
plt.plot(xx, f, 'r-', label='formula')
plt.xlabel('misura (mm)')
plt.ylabel('densita')
plt.title('100000 misure, mu = 25.00 mm, sigma = 0.04 mm')
plt.legend()
plt.savefig('gaussiana.png')
plt.show()
```

